

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 24

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Net held with its straight edge on the river bed facing the current (open to the flow) • The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) • Same substrate type, water depth and flow rate at both sites; same net mesh; repeat samples; invertebrates identified with a key and counted	3
01.2	<i>Any 3 from:</i> • Difficult to standardise the vigour / area kicked • Buried organisms or those fixed to stones are not dislodged • Some organisms swim and avoid the net • Flow rate varies and affects capture • Fragile organisms destroyed	3
01.3	• A fixed area of bed is enclosed by a frame so the sampled area is standardised; side nets reduce escapes • But only works in fast enough flow and is hard to set on rough substrates / deep water	2
01.4	• Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event • Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
02.1	<i>Any 2 from:</i> • Hedgerows • Hay meadows • Grazed moorland / heathland • Chalk grassland	2

Question	Marking guidance	Marks
02.2	• Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive • Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers	2
02.3	• The introduced (non-indigenous) predator may eat other species as well as the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
03.1	• Negative feedback responds to any change by re-establishing the previous equilibrium • Only large unusual events overwhelm the feedback so a new equilibrium forms • Example: methane hydrate release at the end of the Permian causing climate change and mass extinction	3
03.2	• In a diverse system each component is relatively less important so the loss / change of one has a small effect; alternative food sources and relationships buffer the change • Natural: coral reefs and tropical rainforests with complex food webs are stable; simple polar / desert systems fluctuate • Human: agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather • Diverse technical systems: a society using a range of renewable energy resources is more secure than one relying on a single source	4
03.3	<i>Any 3 from:</i> • Melting permafrost releasing methane • Reduced albedo as ice / snow melt • Methane hydrate release • Forest and peat fires • Faster soil decomposition • Ocean acidification reducing sequestration • Cirrus cloud formation	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
04.1	• MEDCs: market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) • LEDCs: herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs; warm climates suit tilapia • Herbivores are more energy-efficient (lower trophic level) so more food per unit area	3
04.2	• Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm • Crossed with normal females (XX) all offspring are XX females • Female trout meat has a better flavour; in tilapia the reverse (all males) because males grow larger / more efficiently	3

Question	Marking guidance	Marks
04.3	• Salmon need high oxygen: water sprays / weirs aerate; wastes removed to prevent deoxygenation • Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density • Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish; controls smoltification • Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	
05.1	• Overall decline from 62% to 22% • Sharp falls in 1998 (to 35%) and 2016 (to 18%) coinciding with bleaching events, with partial recovery after each	2
05.2	• The polyps expel their symbiotic zooxanthellae so the coral loses colour; bleached corals are alive but grow very slowly and may die if algae do not return • Raised water temperature (climate change / El Niño) • Pollution: sewage, pesticides, cyanide, sunscreen; low oxygen from zooplankton blooms after overfishing • High light / solar activity; ocean acidification hinders skeleton formation	4
05.3	• Coastal construction and dredging increase sedimentation / turbidity covering and killing polyps • Sewage and fertiliser nutrients cause algal growth that shades corals and prevents egg release • Physical damage from anchors, divers, souvenir collection; loss of mangroves / seagrass nurseries that trap sediment	3
05.4	• Protected breeding populations grow larger and produce surplus young / adults that spill over into fished areas	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 60-63; AQA spec 3.1.2.3.4.	
06.1	<i>Any 4 from:</i> • Designation of protected areas (SSSIs, MCZs) • Protection of wild birds and their nests (except some pests / game birds) • Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts • Uprooting wild plants generally illegal • Bats must not be disturbed; roof timber treatments must be non-toxic to bats	4
06.2	• Appendix I: species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding • Examples: great apes, big cats, rhinos, blue whale • Appendix II: species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected • Examples: Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap	4
06.3	• Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market • Removing the profit removes the incentive to collect	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.	
07.1	• Light: selective felling to create clearings for plants needing high light while retaining dense canopy for shade species; clearings also give warm sand / soil for lizard eggs; leave dead wood; woodland floor plants (bluebells) grow before the canopy leaves • Dissolved oxygen: narrowing a slow river increases flow and turbulence so more oxygen dissolves, suiting trout and mayfly larvae; shallow water warms quickly for egg development • Nutrients: removing fertile topsoil / not fertilising allows plants of nutrient-poor soils (eg sundew on acidic nitrate-deficient soils) that cannot compete on fertile ground; dung patches add local nutrients	6
07.2	<i>Any 4 from:</i> • Food: wildflower grasslands for seed-eating birds; providing food species • Control of predation: island nest sites unreachable by foxes; trapping / culling / eradicating introduced rats (Scilly Isles - Manx shearwater and storm petrel breeding success increased) • Control of competitors: removing rhododendron so wildflowers get light • Pollination: a range of flowering plants with different flowering times so pollinators have food all season • Seed dispersal species (hippos, elephants, monkeys) conserved; control of introduced pathogens	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 43-46; AQA spec 3.1.2.3.4, 3.1.3.1.	
08.1	• Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins); wastes of one process are raw materials for another so nothing is depleted or accumulates • Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials • Human systems are linear: extract - process - manufacture - use - discard; minerals are dispersed after use or mixed so recovery is difficult • Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up; fossil fuel use is inherently linear and unsustainable	4
08.2	• Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life • They leach from landfill or are released by incineration; persistent, bioaccumulating	2
08.3	• Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) • Easy identification of materials (coding) and separation of components so they can be recycled • Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) • Use of recycled and recyclable / biodegradable materials; reduced packaging and material use (Cradle to Cradle)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	

Question	Marking guidance	Marks
09.1	- Hydrosere - Single-celled algae (and floating plants) - arrive on birds / insects / from soil	2
09.2	- Pioneers and later plants grow and die; dead organic matter plus sediment washed in fill the pond so water becomes shallower - As depth falls rooted plants (pondweeds then emergent reeds with leaves above water) can anchor and colonise; emergents shade submerged plants which die out - Sediment / soil accumulates and becomes able to support larger plants; water-tolerant trees (willow, alder) colonise the waterlogged soil - Transpiration by trees and further sediment dry the soil and make it aerobic so oak can colonise; the denser canopy shades out smaller plants - Each community changes abiotic conditions (depth, light, nutrients, oxygen), making it suitable for the next colonisers which out-compete them; conditions become less extreme and biotic factors increasingly important	5
09.3	- Starting conditions (bare rock vs open water) differ but each stage makes conditions less extreme and builds soil / organic matter - The final community is controlled by the regional climate (temperature, water availability), which is the same for both - climatic climax (eg oak woodland in the UK)	2
09.4	Lichens and algae colonise bare rock (extreme temperatures, no soil, little water); rock fragments and dead organic matter accumulate; mosses, then grasses and ferns as thin soil forms; wind-dispersed trees (birch) then animal-dispersed (oak, beech)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 73-76; AQA spec 3.1.3.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food; new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • Biomimetics: vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples); most species unstudied - argument for habitat conservation • Physiological research using wildlife species; wildlife as pest control agents (ladybirds, Encarsia, Cactoblastis, Bt) • Genetic resources: crop wild relatives, Vavilov centres, gene pools • Ecosystem services: atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments; the unknown value of species not yet investigated (precautionary principle) • Evaluation: which arguments are most persuasive to decision makers; economic valuation vs intrinsic value; judgement	9
10.2	• The variety and abundance of life (species, genes, habitats) in an area / on Earth	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25

Question	Marking guidance	Marks
11.2	Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Temperate broadleaf woodland: seasonality, deep soils, high biodiversity, resources, climate and soil regulation; deforestation, fragmentation, management change; SSSIs, ancient woodland protection, coppicing, mixed planting • Tropical rainforest: stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils; fuelwood, timber, agriculture, mining, reservoirs, climate change; protected areas, debt-for-nature swaps, sustainable exploitation • Tropical coral reefs: cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity; fisheries, erosion protection, medicines, tourism; damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species; MPAs, no-take zones, control of activities • Deep-water reefs: cold, dark, slow growth; research, fisheries; trawling, oil / gas, acidification; SACs • Oceanic islands: isolation, endemism, no mammal predators; exploitation, introduced species, habitat change, sea level; eradication, control of development and visitors • Mangroves: halophytes, anaerobic mud; erosion protection, fisheries nurseries, sediment trapping; clearance for aquaculture / urban / ports, pollution, climate change; protection, replanting • Antarctica: extreme cold, seasonality, marine nutrients, ice; water store, albedo, carbon, research; climate change, ozone, tourism, fishing, minerals; Antarctic Treaty, CCAMLR, tourism and waste rules • Synthesis: features -> importance -> threats -> strategy; evaluation of success; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	25